

Functionality Specification For The Lost & Found Item Tracker

Group 8

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1. Functional Requirements

1.1 Data Requirements

The system must support structured and secure handling of item and user data in order to facilitate core lost and found functionality. The data that must be collected and processed includes item name, description, category, date found/lost, location, item status, and optional image for data relating to items, and user email address for data relating to users. It is important to note that no additional personal data should be stored in accordance with FERPA and data minimization principles recommended under GDPR.

All data captured by the system must be validated prior to storage. Item descriptions must not contain personal identifiable information. Date fields must not accept future dates. Uploaded files must adhere to approved format and size limits. These validation requirements align with the OWASP ASVS recommendations for secure data handling.

1.2 Operations Performed

The system shall support a defined set of operations that enable users to report, search for, and retrieve information about lost and found items in a structured and efficient manner. These operations shall be designed to ensure compliance with applicable privacy, security, and institutional regulations.

The system shall allow users to submit structured reports for either lost or found items. Each report shall include only non-personal information such as item type, description, location, time frame, and the optional media attachments. User-submitted data shall exclude personal information to maintain compliance with FERPA, GDPR, and university standards.

Users shall be able to browse a continuously updated list of reported items. The system shall support filtering by category, date, location, and item status. These operations shall be designed with efficiency and accuracy in mind.

The system shall provide item detail pages containing descriptions supplied by the reporting user. Contact mechanisms shall be designed using a secure message protocol or anonymized communication when applicable.

The system shall allow users to withdraw their submitted item listing. Data removal operations shall comply with data retention requirements applicable under university policy and GDPR, including the right to remove data from the database.

Users shall be permitted to update a previously submitted report. All updates shall be logged in accordance with university record-keeping requirements.

The system shall include the ability for authorized users to remove inappropriate content, flag policy violations, and address misuse as needed. Administrative actions shall be restricted to designated roles and shall follow OWASP access control principles.

1.3 Data Workflows

The system must implement data workflows governing the lifecycle of item information:

1. The item is submitted through a designated form.
2. The data is validated and stored in the database.
3. The item becomes accessible in the searchable item listing.
4. A user may request to claim the item.
5. The system archives the item after resolution or a set expiration time frame.

Because this system is used on the UWM campus, data retention must follow UWM IT governance and GDPR storage limitation principles, which require that item records be archived or deleted when no longer operationally necessary.

1.4 System Outputs

The system must produce the following outputs:

- Public Item Listings: filtered and searchable displays of active items, without displaying user data
- Administrative Reports: summaries of total items submitted, claimed, archived, and active, without displaying student information to maintain FERPA compliance.
- Audit Logs: internal logs documenting submissions, edits, and administrative actions, only made accessible to authorized personnel.
- Communication Logs: provides a record of all interactions between owners and reporters. Include timestamps for each message, the identities of the participants, and content type. Ensures transparency, supports dispute resolution, and maintains an auditable history of all communications related to the item's recovery process.
- Match Notification: Alerts and informs users of lost items that closely correspond to their own. Provides a brief summary of both items. Notification is designed to streamline the recovery process, quickly bringing possible matches to the owner.

1.5 Access Control Requirements

To properly delineate who enters data, the overall users include those that are responsible for reporting lost items, the claimants of lost items, administrators of the software, and potential moderators.

The standard users, provided that they have been authenticated, are allowed to submit a found item, where data including title, description, location, category, images, and private messages are viewed and utilized. Simply, the standard users are granted surface level view of the application, and only the user interface is available for alterations.

Moderators are granted higher permissions in areas that concern the maintenance of the application. Their roles include but aren't limited to answering user reports, reviewing false claims, investigating disputes, accessing flagged records, and moderation of user conflicts. Additional interface to maintain and the social and practical integrity of the application is assigned.

Administrators such as system owners, developers, and employees have access to the codebase and full permissions. Personal data of every user may be viewed and modified, sensitive data concerning the application are given, handling of legal obligations, and all of the above.

2. Interface Requirements

2.1 User Interface Requirements

The system should possess a main page that supports the core set of operations and views. These views must include a clean, responsive, and simple interface with options to allow specific actions such as taking the user to corresponding extended pages. These include searches of lost items based on filters, notifications, submission of items, and communication.

Users should be able to view and respond to chat messages through usual texting styles. The interface should also provide a dashboard showing a user's reported lost and found items and current interactions. The UI must comply with accessibility standards, including readable fonts, hints for images, and a possible screen reader compatibility. Most common image formats are to be supported for uploading and scanning, such as PNG, JPG, and JPEG.

2.1.1 Field Requirements

- Item Name: Accepts text input
- Description: Accepts text input
- Category: Select from a predefined list of categories
- Date Found/Lost: Only accepts dates on or before today
- Location: Accepts text input
- Status: The item's current state (Lost, Found, Claimed, Archived) defaults to Lost
- Item Image: Accepts jpg/png/jpeg file
- Search Field: Accepts text input
- Search Filters: Category, Date (searches dates on or after selected date and before today), Location, Status (won't appear in results if item is marked as claimed)

2.1.2 Screen Requirements

- Home: Displays links to Item Submission, Item Search, and Dashboard screens
- Item Submission: Displays required fields for submitting items (Item Name, Description, Ect.) and confirmation upon item submission
- Item Search: Displays required fields for searching, filtering searches, and a list of items based on search and filter criteria (Clicking item links to its Item Detail screen)
- Dashboard: Displays a list of items associated with the user's claims (Clicking item links to its Item Detail screen), a list of possible item matches based on users' "Lost" item submissions (Clicking item links to its Item Detail screen), and a list

of the user's Lost and Found Submissions (Clicking item allows user to remove or edit item information)

- Item Details: Displays required fields from Item Submission (Item Name, Description, Ect.) and a means to message the finder
- Admin: Only accessible by admin accounts and displays all, flagged, or archived items in order to moderate and remove content

2.1.3 Interaction/Accessibility Requirements

- Required fields must be filled out for completion of Item Submission
- User input fields provide inline error messages for invalid inputs
- Search results must update based on user input and filters

3. Business Requirements

3.1 Operational Rules and Data Dependencies

The system must enforce the following business logic to ensure consistent and compliant operations:

- Item data must be entered and validated before an item becomes visible in public listings.
- An authenticated user must initiate the claim workflow before an item can be marked as resolved.
- Administrators must be able to archive items that have been claimed or that have exceeded a designated retention period.
- The system must adhere to FERPA by ensuring no student-identifying information is displayed publicly or shared outside authorized workflows.
- The system must adhere to GDPR-informed principles by ensuring the minimization, purpose limitation, and timely deletion of stored user data.

4. Security Requirements

4.1 Client-Server Data Handling

The system shall utilize strategies to prevent common attack vectors:

- UI and database actions that have permission requirements will always have those requirements checked by the server
- Using strictly defined “prepared statements” in database interaction scripting to prevent SQL injection attacks
- All cross-interface variables will be sanitized prior to server-side handling to prevent XSS attacks

4.2 General Practice

The system shall apply a range of strategies recommended by OWASP:

- Only showing detailed error descriptions to users with Moderator-level permissions by using a generalized upper-layer of error handling
- Using HTTPS to encrypt all traffic between client and server
- Only asking for and storing user information crucial for the service's operation
 - An example of unnecessary information include a user's birthday
- Regular manual updates of dependencies aided by automatic checking
- Using test-cases meant to explicitly test security features that enforce, among others, the principle of least privilege
- Keeping a centralized log of activity to both prevent and learn from attacks

Citations

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